

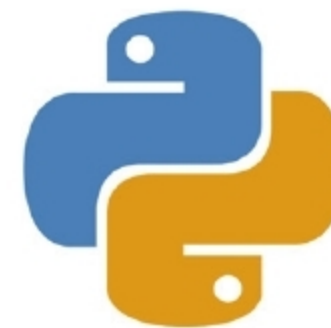
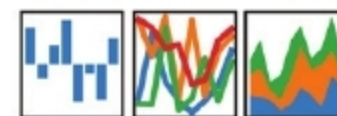
Visualisation Versus Summary Statistics: The Lesson from Anscombe's Quartet

When comparing two (or more) data sets, the statistics about each may not give the entire picture. Often, the data has to be actually plotted or visualised to get the true picture. This article uses two Python libraries — Pandas (for data downloading and manipulation) and Matplotlib — for plotting. The script is in five parts to make it easier to understand.



pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$



Unless you have done some formal/informal courses in statistics or data visualisation, the chances are that you must have never heard of Anscombe's quartet.

This article is about three things:

- Visualisation
- Summary statistics
- Anscombe's quartet

Let's begin with Anscombe's quartet. According to Wikipedia:

"Anscombe's quartet comprises four data sets that have nearly identical simple descriptive statistics, yet have very different distributions and appear very different when graphed. Each data set consists of eleven (x, y) points."

These data sets were constructed in 1973 by the statistician Francis Anscombe to demonstrate both the importance of graphing data before analysing it as well as the effect of outliers and other influential observations on statistical properties.

To summarise:

- Anscombe's quartet comprises four data sets in the form of x-y pairs. (Because the number of data pairs is four, hence the term quartet.)
- Each data set has eleven points.
- The 'summary statistics' of all the four data sets are near equal or identical.
- But when these four data sets are plotted, they represent altogether different graphs.

Before discussing the data sets, let us first understand the term 'summary statistics'. We won't get into the semantics of the term. We just need to understand that when we compare two data sets 'quantitatively', we compare them on parameters like:

- Mean (which is a measure of central tendency)
- Variance (which is a measure of spread)
- Correlation (which is a measure of dependence)
- Linear regression (which indicates the best fit line)
- Coefficient of determination (which indicates the proportion of variance in the dependent variable that is predicted from the independent variable)

